

PHYSICI 101 LAB: PHYSICAL SCIENCE LABORATORY INVESTIGATIONS



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Physical Science 101 Lab

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TABLE OF CONTENTS

Licensing

1: The Scientific Method

- 1.1: Interpreting Errors

2: Investigation 1 - Making a Hypothesis

- 2.1: Materials
- 2.2: Introduction
- 2.3: Pre-lab
- 2.4: Procedure
- 2.5: Analysis
- 2.6: General Question

3: Investigation 2 - Precision and Accuracy

- 3.1: Materials
- 3.2: Introduction
- 3.3: Pre-lab
- 3.4: Procedures
- 3.5: Analysis
- 3.6: General Questions

4: Investigation 3 - The Best Paper Towel

- 4.1: Materials
- 4.2: Introduction
- 4.3: Pre-lab
- 4.4: Procedures
- 4.5: Analysis
- 4.6: General Questions

5: Investigation 4 - Measuring a Wood Block

- 5.1: Materials
- 5.2: Introduction
- 5.3: Pre-Lab
- 5.4: Procedures
- 5.5: Analysis
- 5.6: General Questions

6: Investigation 5 - Walking at a Constant Speed

- 6.1: Materials
- 6.2: Introduction
- 6.3: Pre-Lab
- 6.4: Procedures
- 6.5: Analysis
- 6.6: General Questions

7: Investigation 6 - Falling Objects

- 7.1: Materials
- 7.2: Introduction
- 7.3: Pre-Lab
- 7.4: Procedures
- 7.5: Analysis
- 7.6: General Questions

8: Investigation 7 - Traveling in a Circle

- 8.1: Materials
- 8.2: Introduction
- 8.3: Pre-Lab
- 8.4: Procedures
- 8.5: Analysis
- 8.6: General Questions

9: Investigation 8 - Firing at a Distant Target

- 9.1: Materials
- 9.2: Introduction
- 9.3: Pre-Lab
- 9.4: Procedures
- 9.5: Analysis
- 9.6: General Questions

10: Investigation 9 - Balanced and Unbalanced Forces

- 10.1: Materials
- 10.2: Introduction
- 10.3: Procedures
- 10.4: Analysis
- 10.5: General Questions

11: Investigation 10 - Weight of Mass

- 11.1: Materials
- 11.2: Introduction
- 11.3: Pre-Lab
- 11.4: Procedures
- 11.5: Analysis
- 11.6: General Questions

12: Investigation 11 - High Flying Bottle Rocket

- 12.1: Materials
- 12.2: Introduction
- 12.3: Pre-Lab
- 12.4: Procedures
- 12.5: Analysis
- 12.6: General Questions

13: Investigation 12 - The Pressure on Your Feet

- 13.1: Materials
- 13.2: Introduction
- 13.3: Pre-Lab
- 13.4: Procedures
- 13.5: Analysis
- 13.6: General Questions

14: Investigation 13 - Mass vs Volume vs Density

- 14.1: Materials
- 14.2: Introduction
- 14.3: Pre-Lab
- 14.4: Procedures
- 14.5: Analysis
- 14.6: General Questions

15: Investigation 14 - Sinking and Floating

- 15.1: Materials
- 15.2: Introduction
- 15.3: Pre-Lab
- 15.4: Procedures
- 15.5: Analysis
- 15.6: General Questions

16: Investigation 15 - Two Wood Rafts

- 16.1: Materials
- 16.2: Introduction
- 16.3: Pre-Lab
- 16.4: Procedures
- 16.5: Analysis
- 16.6: General Questions

17: Investigation 16 - Magic Coins and Paperclips

- 17.1: Materials
- 17.2: Introduction
- 17.3: Pre-Lab
- 17.4: Procedures
- 17.5: Analysis
- 17.6: General Questions

18: Investigation 17 - Weight Lifting

- 18.1: Materials
- 18.2: Introduction
- 18.3: Pre-Lab
- 18.4: Procedures
- 18.5: Analysis
- 18.6: General Questions

19: Investigation 18 - Rolling Down a Hill

- 19.1: Materials
- 19.2: Introduction
- 19.3: Pre-Lab
- 19.4: Procedures
- 19.5: Analysis
- 19.6: General Questions

20: Investigation 19 -The Swing

- 20.1: Materials
- 20.2: Introduction
- 20.3: Pre-Lab
- 20.4: Procedures
- 20.5: Analysis
- 20.6: General Questions

21: Investigation 20 - Balancing a See-Saw

- 21.1: Materials
- 21.2: Introduction
- 21.3: Pre-Lab
- 21.4: Procedures
- 21.5: Analysis
- 21.6: General Questions

22: Investigation 21 – The Temperature of a Molecule

- 22.1: Materials
- 22.2: Introduction
- 22.3: Pre-Lab
- 22.4: Procedures
- 22.5: Analysis
- 22.6: General Questions

23: Investigation 22 - Heat Energy Transfer

- 23.1: Materials
- 23.2: Introduction
- 23.3: Pre-Lab
- 23.4: Procedures
- 23.5: Analysis
- 23.6: General Questions

24: Investigation 23 - Rates of Heat Conduction

- 24.1: Materials
- 24.2: Introduction
- 24.3: Pre-Lab
- 24.4: Procedures
- 24.5: Analysis
- 24.6: General Questions

25: Investigation 24 - Phase Change of Water vs Salt Water

- 25.1: Materials
- 25.2: Introduction
- 25.3: Pre-Lab
- 25.4: Procedures
- 25.5: Analysis
- 25.6: General Questions

26: Investigation 25 - Magically Moving Objects

- 26.1: Materials
- 26.2: Introduction
- 26.3: Pre-Lab
- 26.4: Procedures
- 26.5: Analysis
- 26.6: General Questions

27: Investigation 26 - Holiday Lights

- 27.1: Materials
- 27.2: Introduction
- 27.3: Pre-Lab
- 27.4: Procedures
- 27.5: Analysis
- 27.6: General Questions

28: Investigation 27 - Magnetic Field Patterns and Interactions

- 28.1: Materials
- 28.2: Introduction
- 28.3: Pre-Lab
- 28.4: Procedures
- 28.5: Analysis
- 28.6: General Questions

29: Investigation 28 – The Electromagnetic Connection

- 29.1: Materials
- 29.2: Introduction
- 29.3: Pre-Lab
- 29.4: Procedures
- 29.5: Analysis
- 29.6: General Questions

30: Investigation 29 - Energy of Sound and Speed of Light

- 30.1: Materials
- 30.2: Introduction
- 30.3: Pre-Lab
- 30.4: Procedures
- 30.5: Analysis
- 30.6: General Questions

31: Investigation 30 - Waves on a Very Long Spring

- 31.1: Materials
- 31.2: Introduction
- 31.3: Pre-Lab
- 31.4: Procedures
- 31.5: Analysis
- 31.6: General Questions

32: Investigation 31 - Bouncing Light

- 32.1: Materials
- 32.2: Introduction
- 32.3: Pre-Lab
- 32.4: Procedures
- 32.5: Analysis
- 32.6: General Questions

33: Investigation 32 - Bending Light

- 33.1: Materials
- 33.2: Introduction
- 33.3: Pre-Lab
- 33.4: Procedures
- 33.5: Analysis
- 33.6: General Questions

34: Investigation 33 - The Atomic Spectra of Neon Lights

- 34.1: Materials
- 34.2: Introduction
- 34.3: Pre-Lab
- 34.4: Procedures
- 34.5: Analysis
- 34.6: General Questions

35: Investigation 34 - Modeling Radioactive Half-Life

- 35.1: Materials
- 35.2: Introduction
- 35.3: Pre-Lab
- 35.4: Procedures
- 35.5: Analysis
- 35.6: General Questions

36: Investigation 35 - Identifying Chemical Changes

- 36.1: Materials
- 36.2: Introduction
- 36.3: Pre-Lab
- 36.4: Procedures
- 36.5: Analysis
- 36.6: General Questions

37: Investigation 36 - Mixing Solids and Liquids

- 37.1: Materials
- 37.2: Introduction
- 37.3: Pre-Lab
- 37.4: Procedures
- 37.5: Analysis
- 37.6: General Questions

38: Investigation 37 - Pattern in the Periodic Table

- 38.1: Materials
- 38.2: Introduction
- 38.3: Pre-Lab
- 38.4: Procedures
- 38.5: Analysis
- 38.6: General Questions

39: Investigation 38 - Building Models of Molecules

- 39.1: Materials
- 39.2: Introduction
- 39.3: Pre-Lab
- 39.4: Procedures
- 39.5: Analysis
- 39.6: General Questions

40: Investigation 39 - Melting Wax and Cooking Sugar

- 40.1: Materials
- 40.2: Introduction
- 40.3: Pre-Lab
- 40.4: Procedures
- 40.5: Analysis
- 40.6: General Questions

41: Investigation 40 - Dozens and Moles

- 41.1: Materials
- 41.2: Introduction
- 41.3: Pre-Lab
- 41.4: Procedures
- 41.5: Analysis
- 41.6: General Questions

42: Investigation 41 - Rates of Reactions

- 42.1: Materials
- 42.2: Introduction
- 42.3: Pre-Lab
- 42.4: Procedures
- 42.5: Analysis
- 42.6: General Questions

43: Investigation 42 - Magic Colors

- 43.1: Materials
- 43.2: Introduction
- 43.3: Pre-Lab
- 43.4: Procedures
- 43.5: Analysis
- 43.6: General Questions

44: Investigation 43 - Acids and Bases around the House

- 44.1: Materials
- 44.2: Introduction
- 44.3: Pre-Lab
- 44.4: Procedures
- 44.5: Analysis
- 44.6: General Questions

45: Investigation 44 - Chemistry in a Ziploc Bag

- 45.1: Materials
- 45.2: Introduction
- 45.3: Pre-Lab
- 45.4: Procedures
- 45.5: Analysis
- 45.6: General Questions

46: Investigation 45 - Fruit and Veggie Current

- 46.1: Materials
- 46.2: Introduction
- 46.3: Pre-Lab
- 46.4: Procedures
- 46.5: Analysis
- 46.6: General Questions

47: Appendix I - The Physical Science Tool Box of Units

- 47.1: Materials
- 47.2: Introduction
- 47.3: Procedures
- 47.4: Analysis
- 47.5: General Questions

48: Appendix II - Constructing and Interpreting a Graph

- 48.1: Materials
- 48.2: Introduction
- 48.3: Procedures
- 48.4: Analysis
- 48.5: General Questions

49: Appendix III - Manipulating Numbers in Scientific Notation

- [49.1: Materials](#)
- [49.2: Introduction](#)
- [49.3: Procedures](#)
- [49.5: Analysis](#)
- [49.6: General Questions](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

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CHAPTER OVERVIEW

1: The Scientific Method

The scientific method has been described in numerous textbooks as a set of steps to be followed. Steps may be memorized without understanding the true nature of the steps, the true nature of doing science. The way in which a scientist approaches answering an unknown is through informed experimentation. Scientists approach the unknown as a question to be answered and devise a method, based on prior knowledge and experience, to answer the question. The method relies on precise measurements which allows for the results to be verified. Because scientists are dealing with an unknown, something for which there is not an answer, they must often begin with a hypothesis. A hypothesis is what the informed scientists thinks the answer to the question will be; it is essentially an educated guess.

- Teresa Ciardi

[1.1: Interpreting Errors](#)

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1.1: Interpreting Errors

Systematic Error

- Measurements deviate from the correct value by the same amount each time
- Errors have a pattern

Example: Time on a watch is always 5 minutes ahead of the true time

Random Error

- Errors have no pattern

Example: Everyone's watch shows a different time

Uncertainty in Measurement

- Number $\pm$ error
- It is the amount a number may be off from the correct value

Example: Age ± 365 days

Ratio Comparison

- Shows how two numbers compare
- Reduced fraction shows comparison

Example:

$$\frac{65\text{mph}}{30\text{mph}} = \frac{2.2}{1} \text{ (Means one speed is 2.2 times faster than the other)}$$

Percent Error

- Percentage measurements are off from standard/correct value
- Absolute value

$$\text{Percent Error} = \frac{\text{Experimental/Measured Value} - \text{Accepted/Theoretical Value}}{\text{Accepted/Theoretical Value}} \times 100\%$$

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CHAPTER OVERVIEW

2: Investigation 1 - Making a Hypothesis

Learning Objectives

- Experience making a hypothesis, taking good measurements, and developing a procedure to solve a problem.

[2.1: Materials](#)

[2.2: Introduction](#)

[2.3: Pre-lab](#)

[2.4: Procedure](#)

[2.5: Analysis](#)

[2.6: General Question](#)

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